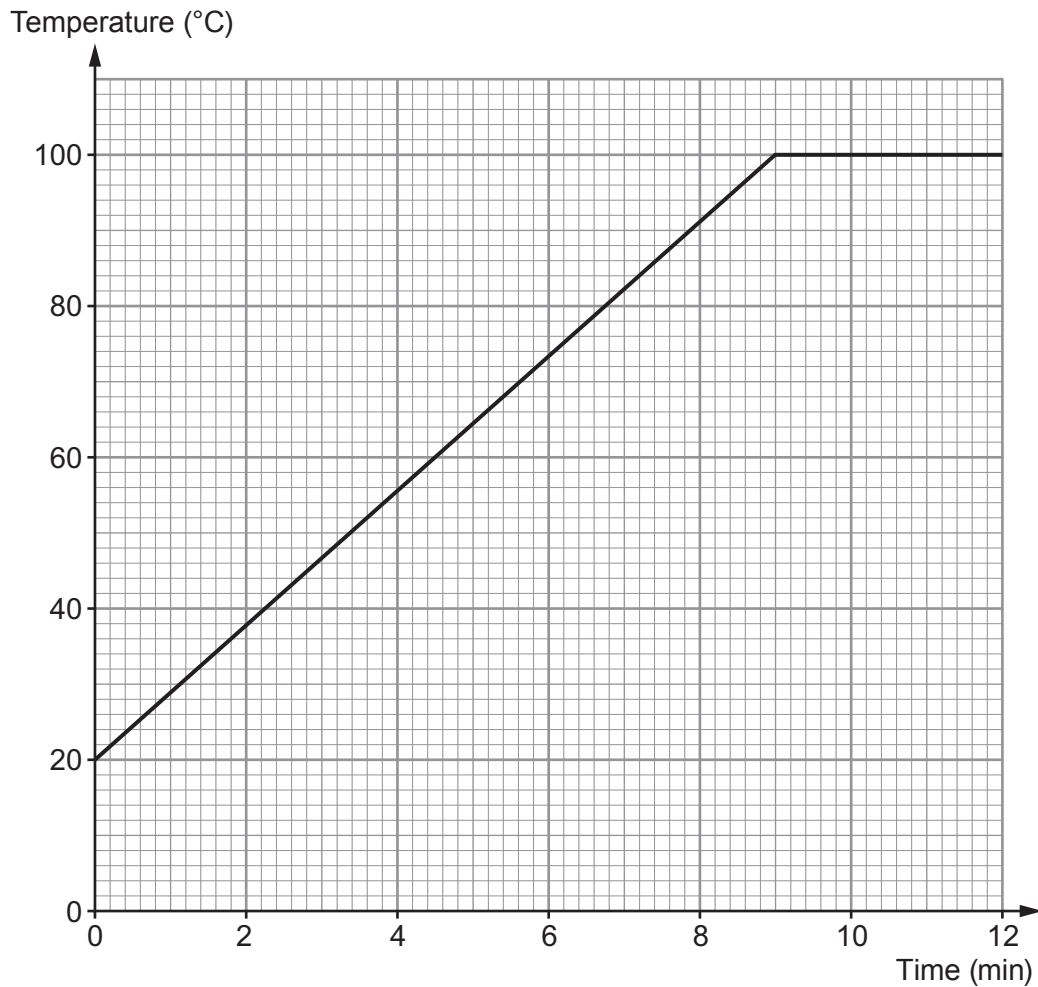


5. The graph below shows how the temperature of 0.2 kg of water changes as it is heated from 20 °C.



- (a) Describe the relationship between temperature and time in the first 9 minutes. [2]

.....

.....

- (b) (i) Write down the temperature of the water at 9 minutes. [1]

Temperature = °C

- (ii) Calculate the change in temperature in the first 9 minutes. [1]

Change in temperature = °C



- (iii) The specific heat capacity of water is 4 200 J/kg °C.
Use the equation:

$$\begin{array}{ccccccc} \text{thermal} & = & \text{mass} & \times & \text{specific heat} & \times & \text{change in} \\ \text{energy} & & & & \text{capacity} & & \text{temperature} \end{array}$$

to calculate how much thermal energy is supplied to the 0.2 kg water in the first 9 minutes. [2]

Thermal energy supplied = J

- (c) Between 9 and 12 minutes the water is boiling and its temperature stays constant even though heat energy is still being supplied. State what is happening to the water during this time. [1]

.....

7

